

University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science & Engineering

Program: B.Sc. in CSE

Mid Term Examination, Autumn 2025

Course Title: Data Structures and Algorithms I

Course Code: CSE 0613211

Marks: 20

Time: 1(one) Hours

(Answer all questions)

Marks

1. a) A hospital maintains patient data in a 2D array Patient [100][12], where each **row** represents a patient and each **column** represents the record of a month (January–December). [04]
Find the address of the record of the **25th patient in June** (assume each record takes 8 bytes and the base address = 5000).
b) A teacher gives her students an array of sorted numbers: A = [10, 15, 25, 40, 55, 60, 70, 75, 80, 90, 100] [04]
She asks them to **find the number 75 using Binary Search**. Show each step of the search process until the number is found.
c) Write the best case, worst case and average case complexities of linear search algorithm. [02]
2. a) Figure 1 pictures a linked list in memory where each node of the list contains a single character. Obtain the actual list of characters using the traversing algorithm of linked list. [02]

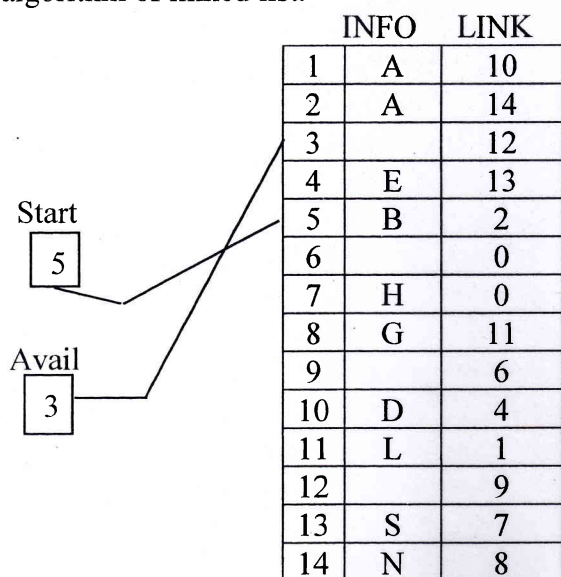
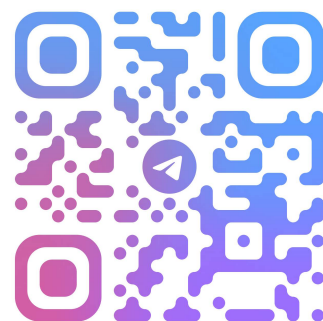


Figure: 1



b) A university maintains student records (name, roll, GPA). New students may join at any time and old records may be removed. [04]

- Explain why a **linked list** is a better choice than an array in this case.
- Show how you would insert a new student record at the beginning.

c) Imagine you are using a **music player app** on your phone. [04]

- The app lets you play songs in a **playlist**.
- You can tap **Next** to go to the following song.
- You can also tap **Previous** to return to the earlier song.

Now the developer has to decide what type of linked list should be used to represent this playlist.

Justify which type of linked list is most appropriate for this scenario and why?

